Contactor and relays for electric drive

Relays and contactors are widely used in industrial automation applications.. Both the devices act like switches and both of them use the principles of electromagnetic induction to function. But does this mean that relays and contactors can be used interchangeably? No, is the short answer.

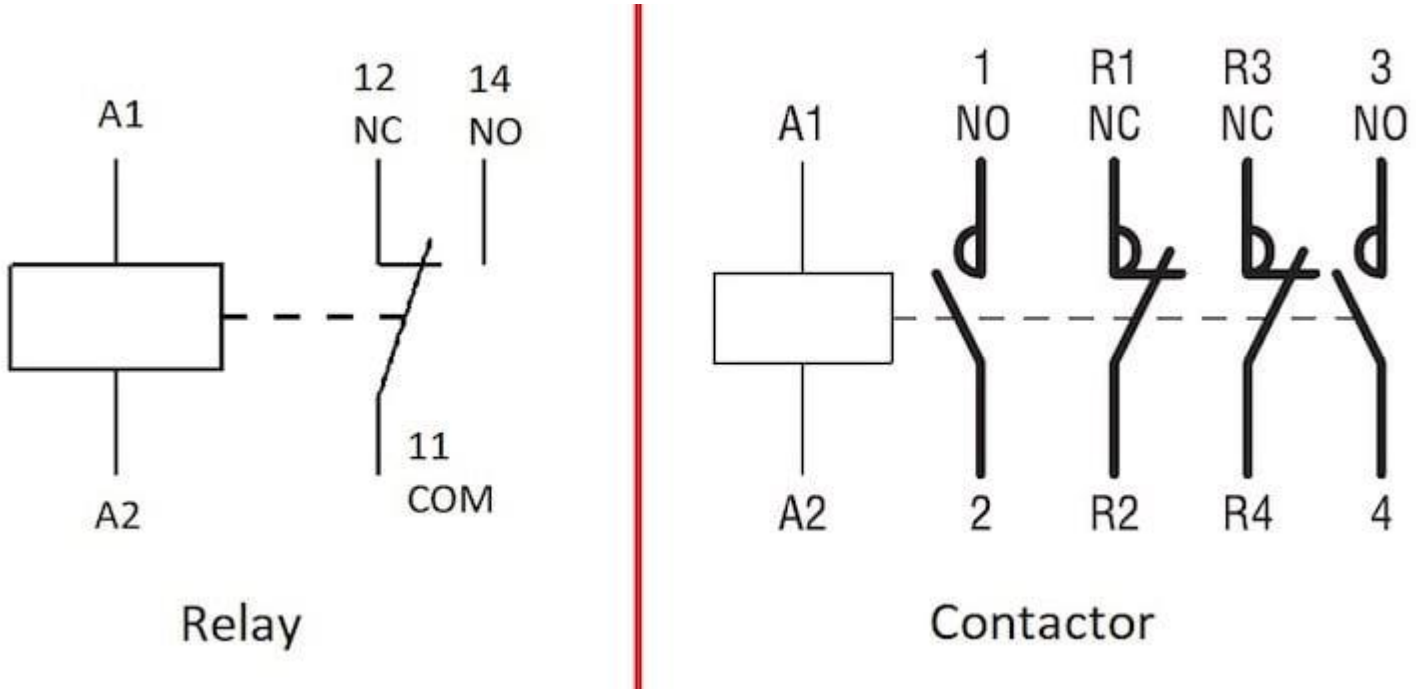
In terms of operation, relays and contactors actually are the same. At the core, relays and contactors are used to separate the control circuit from the main circuit. But functionally, relays and contactors are used in industrial circuits for different specific applications. They cannot be used interchangeably.

– We use a contactor to turn on and off heavy and high voltage electrical devices such as motors, fans, pumps, etc.

– The reason that we use a contactor is to control these heavy high voltage electrical devices indirectly and safely via a PLC and not to connect the PLC directly to these output devices.

– The main difference between a contactor and a relay is that a contactor is used for turning on and off heavy high-voltage devices, but a relay is usually used to turn on and off smaller low-voltage devices.

Contactor and relays for electric drive



relays

Both electromagnetic and solid state (electronic) relays are used. Electromagnetic relay coils have typical ratings of 12 V, 24 V, 48 V dc and output contact ratings up to 240 V, 5 A typically. Relays could have up to 4 N/O (normally open), 4 NC (normally closed) and 4 change over contacts. The N/O, N/C and changeover contacts are shown in the figure below.

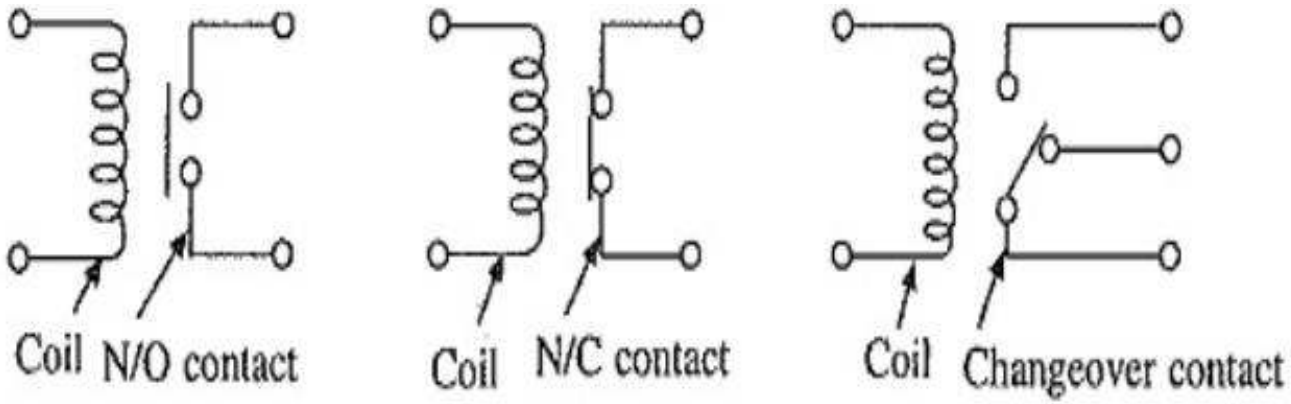
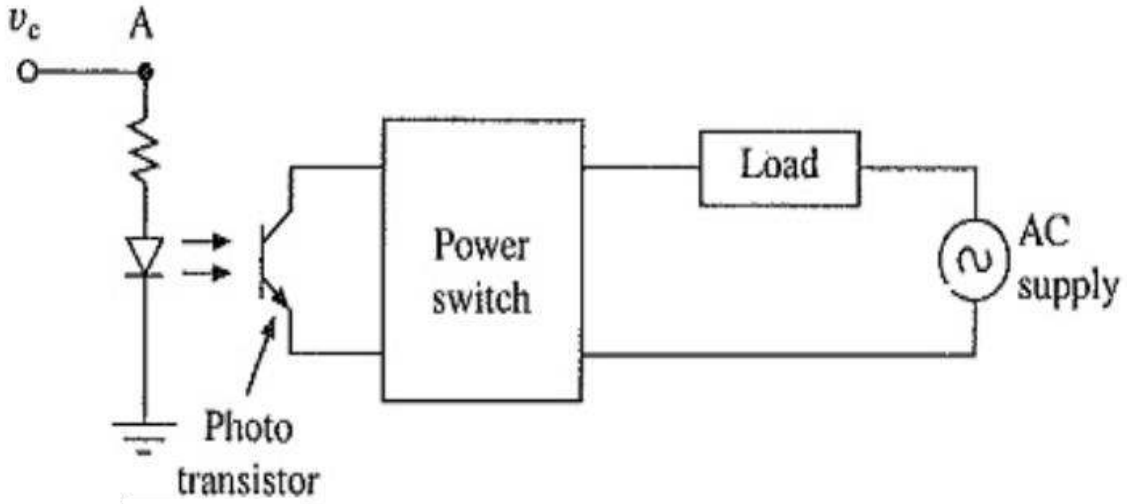


Fig. N/O (normally open), N/C (normally closed) and changeover contacts

Solid state relay : In a solid state relay, the relay function is achieved by means of electronic Interlocking Components in Electrical Drives. It also has isolated input and output, but no moving contacts. Functional diagram of solid state relay is shown in the figure below



Functional diagram of a solid state relay

When control signal is applied across terminal A,LED is turned on.

Light from LED turns on a switch, usually phototransistor or light-activated SCR. With the help of a zero voltage switch, the relay turns on and applies power to the load at the instant of zero crossing of the source voltage. When control signal is removed, power to the source is disconnected when the load current passes through zero. Turning on and off of power to the load at zero current reduces the generation of electromagnetic interference(EMI), which is a problem with the electromagnetic relays. Solid-state relay has several other advantages over electromagnetic relays due to the absence of moving parts such as longer life, maintenance-free operation, and resistance to shock and vibration. Whereas EM relays have multiple contacts(NO, NC, and changeover). Also, they have the ability to withstand surges as those countered in motor starting and braking and resistance to triggering by transient voltages.

Contactors

- A relay is employed in control circuits whereas a contactor is high-power relay which has heavy-duty contacts designed to close and break power circuits. The ratings of contactors are:
- Low-current contractors for interlocking and sequencing: coli voltage 240V, output contact ratings 240V,40A.
- Power contactors: coil voltage 240V, output contact ratings 660V,880A and auxiliary contact ratings up to 10A.
- Bar mounted contactors: coil voltage 240V, output contact ratings 1000V, 1200A and auxiliary contact rating up to 10A.

Auxiliary Contacts: Contactors and circuit breakers usually have few auxiliary contacts (up to a maximum 4 N/O and 4 N/C), in addition to the main (or power) contact. Auxiliary contacts are low current, low voltage contacts similar to relays (generally rated up to 240 V, 5 A), and are mainly employed to sense the state of breaker or contactor.

Timers

- If a particular drive requirement is to be met in a given time, a timer is used in the control logic. Timer can be pneumatic, electromechanical or electronic. Time delay relays are examples of timers and their setting is done based on the need. Its output contacts change from N/O to N/C or vice versa when the time setting is exceeded. Output contacts are available in suitable combinations such as 2 N/O and 2 N/C or 4 changeover. Ratings are 240 V, 5 A.

Circuit Breaker and fuse

Function of the overcurrent protective device is to protect the motor, power modulator and other devices in the power circuit from short circuits and grounds. The two devices commonly employed are circuit breaker and fuse. Both these devices should be capable of carrying motor currents during normal transient operations such as starting, braking and reversing without disconnecting motor from the line. For a motor fed from power modulator (or converter), these devices are set to match the transient rating of power modulator or the motor, whichever has lower value.

When motor is fed directly, overcurrent setting is made between 250 and 400% of full load. Overcurrent protection not only provides overcurrent protection, but also disconnects motor from the source. The circuit breaker accomplishes both functions: When fuses are employed for overcurrent protection, a switch for disconnecting motor from the supply is also provided.

Overload protection

- The overload relay protects the motor from torque overloads, Under voltage, unbalancing voltage, and single phasing. Maximum overload occurs when the motor shaft is stalled (locked rotor condition). The higher the overload, the lesser the time taken by winding temperature to cross the safe limit (curve A).
- Overload relay should have a matching relationship between current overload and the time required to trip and should lie below curve A. This relation is shown by curve B.
- There are two types of overload relays: thermal and magnetic. Bimetallic thermal overload relays which are commonly used have a heating coil which carries the motor winding current. Heat produced by heating coil heats up bimetallic strip. The unequal expansion of two metals makes the strip to bend. Strip pushes against the breaker contacts, causing the contacts to open. The magnetic overload relay has a movable core inside a coil through which the motor current flows. Normal motor current is not enough to cause core to move. When overload occurs, the core moves'. The movement is opposed and damped by a thick viscous liquid or dashpot. The relay trips after certain time depending on the overload (curve B).

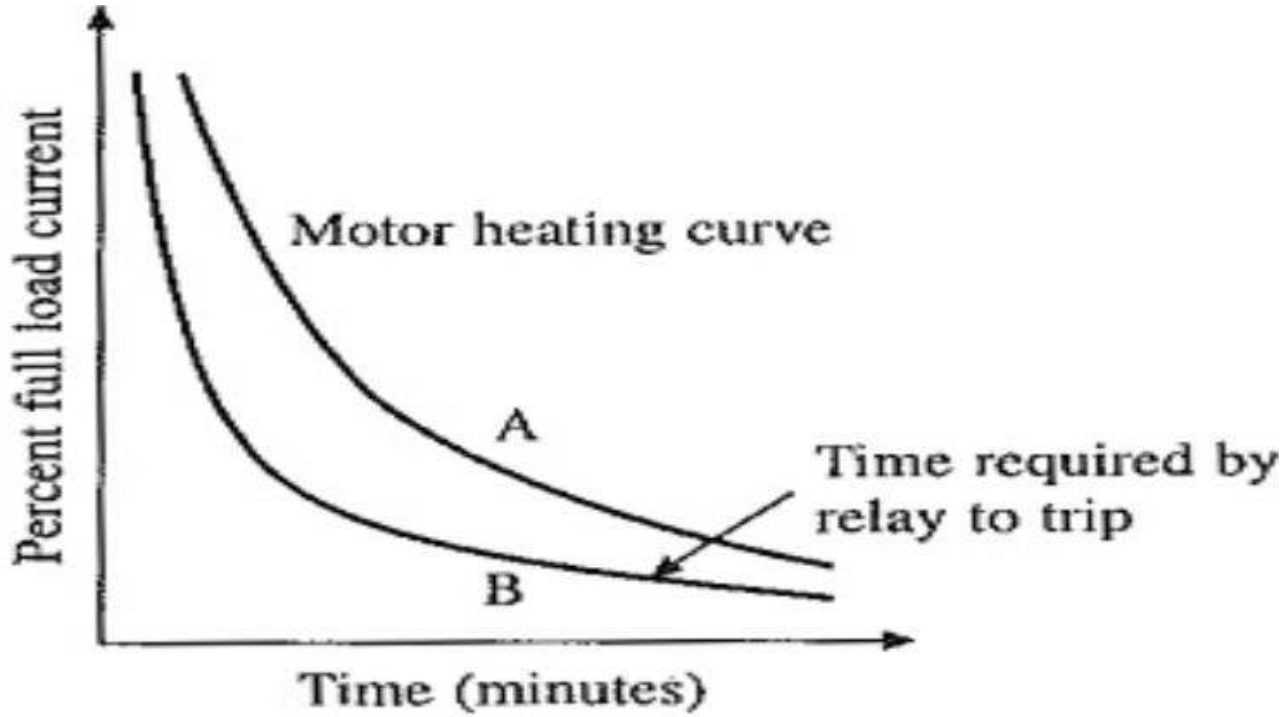


Fig. Setting of overload relay characteristics

Ground fault protection

Large and expensive motors are usually provided with ground fault protection. Ground fault occurs when a current carrying conductor accidentally touches a grounded conducting material. Depending on the material resistance, the current may be small or large. A large current will usually blow a fuse or trip a circuit breaker whereas small current may not be enough to do so. Because of high resistance between conducting material and the ground, potential of the conducting material can be large enough to kill or injure humans. The fault current, if allowed to flow for large periods, can cause fire.

- The ground fault relay works on the principle that in an ac motor, the sum of three-phase current during fault will not be zero, and in a dc motor the sum of currents of incoming and outgoing leads will not be zero.

Contactor and relays for electric drive

Electric motors start, stop and reverse the direction of rotation automatically according to the following principles:

1 -Time Principle
2- Speed Principle
3- Current or Voltage Principle
4- The Displacement (Distance) Principle

Types of time relays according to the principle of operation

A- Electromagnetic timers

- It is commonly used in electrical control circuits. And its installation, as shown in Figure (8-1), is a coil whose ends are connected to the feeding source, and a core moves forward and backwards connected to the holder and on it the moving
contacts are fixed, either they are Normally open (NO) contacts (1,2) or Normally Closed contacts (NC) (3,4).

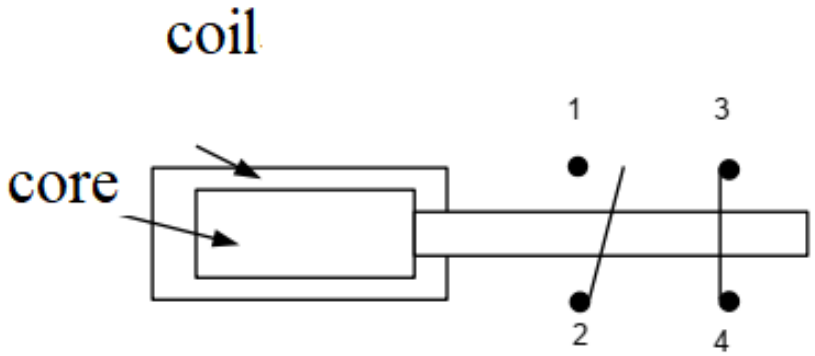


Figure (8-1) Electromagnetic time relay

Principle of operation:

- When the relay coil is connected to the source, a magnetic field is formed that leads to the movement of the metal core and the contacts holder, so that the position of the contacts changes- (1,2) or Normally Closed contacts (NC) (3,4).
- The time delay is achieved in several ways, including connecting a resistance and a capacitor in parallel with the coil (Figure 8-2), where the time of discharge of the capacitor charge (time delay) is controlled by means of changing the value of the resistance.

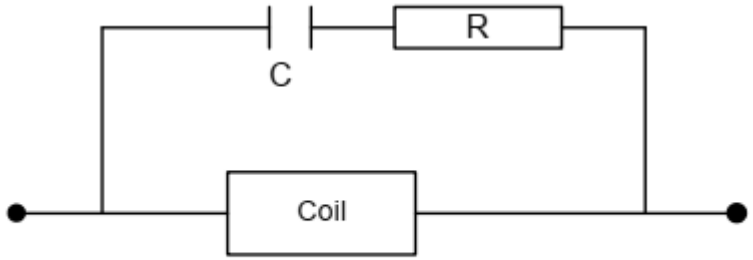


Figure (8-2) Tme delay circuit

B- Electronic timer relay

An electronic relay uses semiconductors, especially transistors. Figure (3-8) shows an electronic time relay

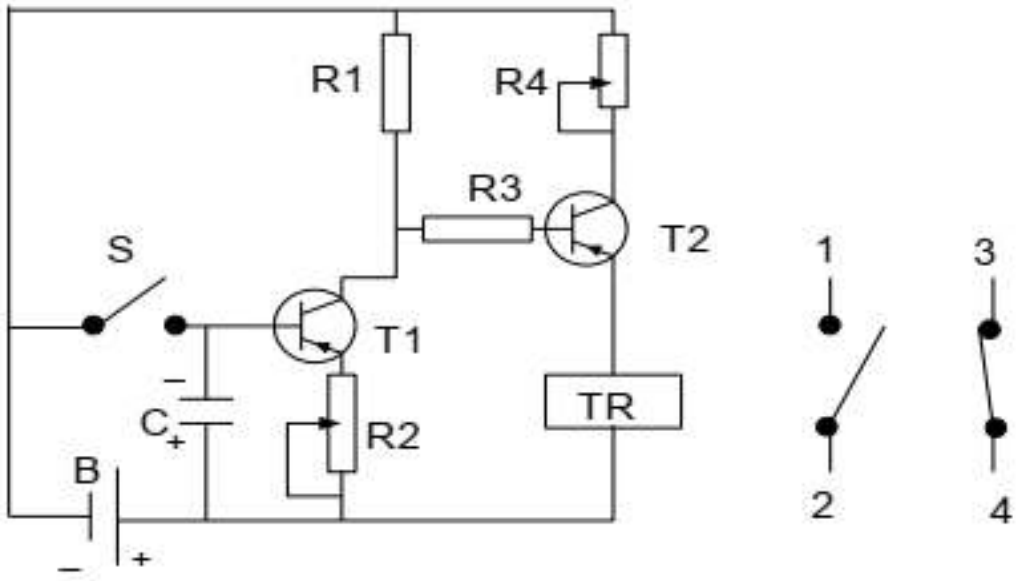


Figure (3-8) Electronic Time Relay

Principle of operation:

- When the switch (S) is closed, the capacitor (C) is charged with the charge shown in Fig. (8-3) and the transistor (T_1) is open(on) and the transistor (T_2) is closed(off). That is, it does not pass a current into the relay (TR) coil. And its contacts are in its normal position.
  When the switch (S) is opened, the capacitor charge (C) is discharged through the resistance (R2) and the transistor (T_1), which causes the transistor (T_2) to open(on) so that current passes from the source through the time relay coil (TR), the transistor (T_2) and the resistor (R3), so its contact position changes.
  The time delay can be calibrated by choosing the value of (R_2)
  where $T = R_2 C$

C-Software time relays

It is used by programming with a Microprocessor, Microcontroller or Program Logic Controller (PLC), and it is easy to use as it is connected and programmed by the user's control program.

Types of time relay according to the origin status of their contacts

- The time relays are divided in terms of time delay into two types: -
  1- Time delay ON Figure (8-4) shows the symbols of the time delay for this type. The status of the relay contacts changes after a period of feeding the relay coil, and they return directly to normal position when the relay coil is disconnected from the source as shown in the timing diagram in Figure (8-5).
- 2- Time Delay OFF Timer, Figure (8-4) shows the symbols of the time delay for this type. The status of the relay contacts changes directly upon the connection of the source to the relay coil and returns to the normal position after a period of disconnecting of the source from the relay coil as shown in the timing diagram in
  Figure (8-5).

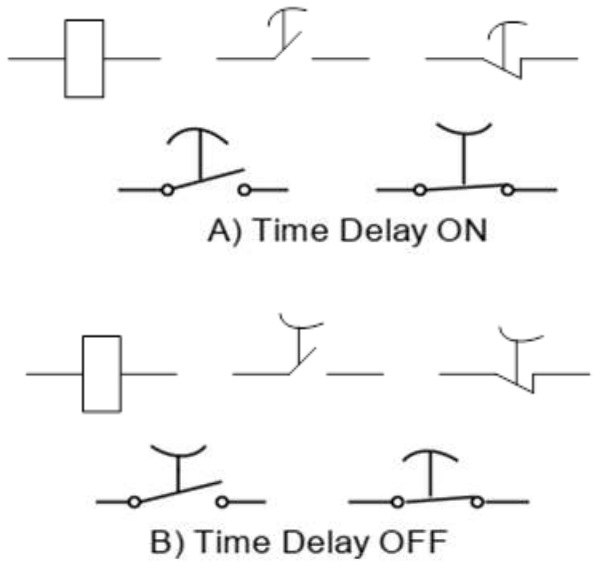


Figure (8-4) Symbol for types of time delay

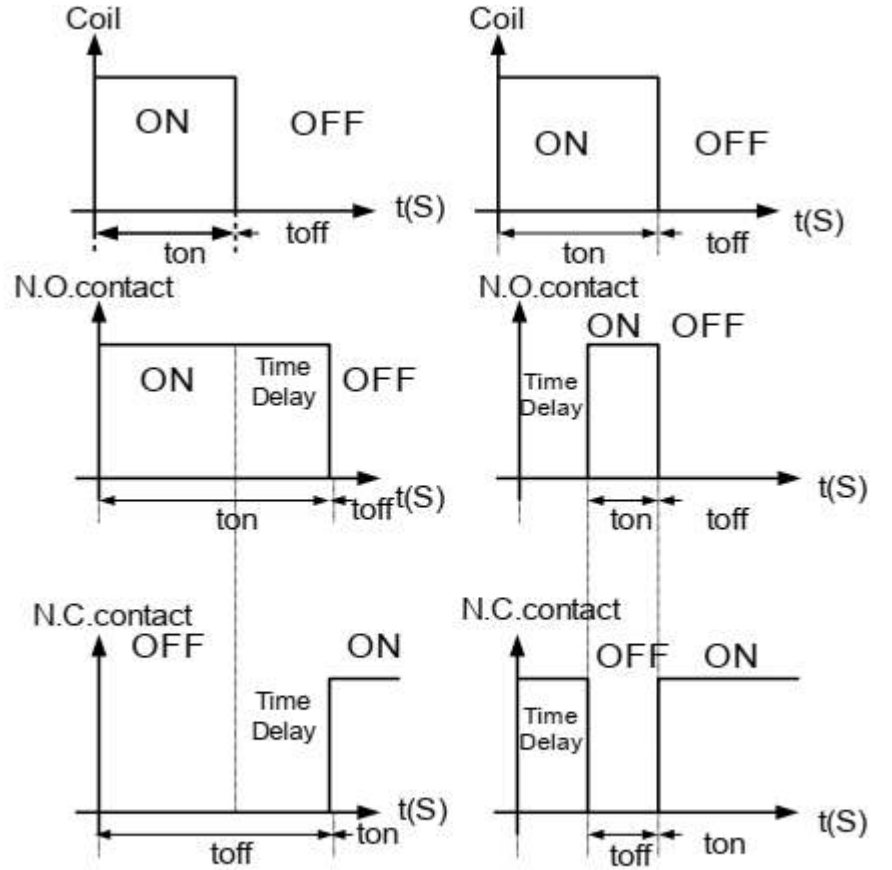


Figure (8-5) Time delay timing diagram

Control circuits for automatic starters of DC and AC motors

A starter is a control device that is used for switching the motor either manually or automatically. It is used for safe ON/OFF control of electrical motors by making or breaking its contacts.

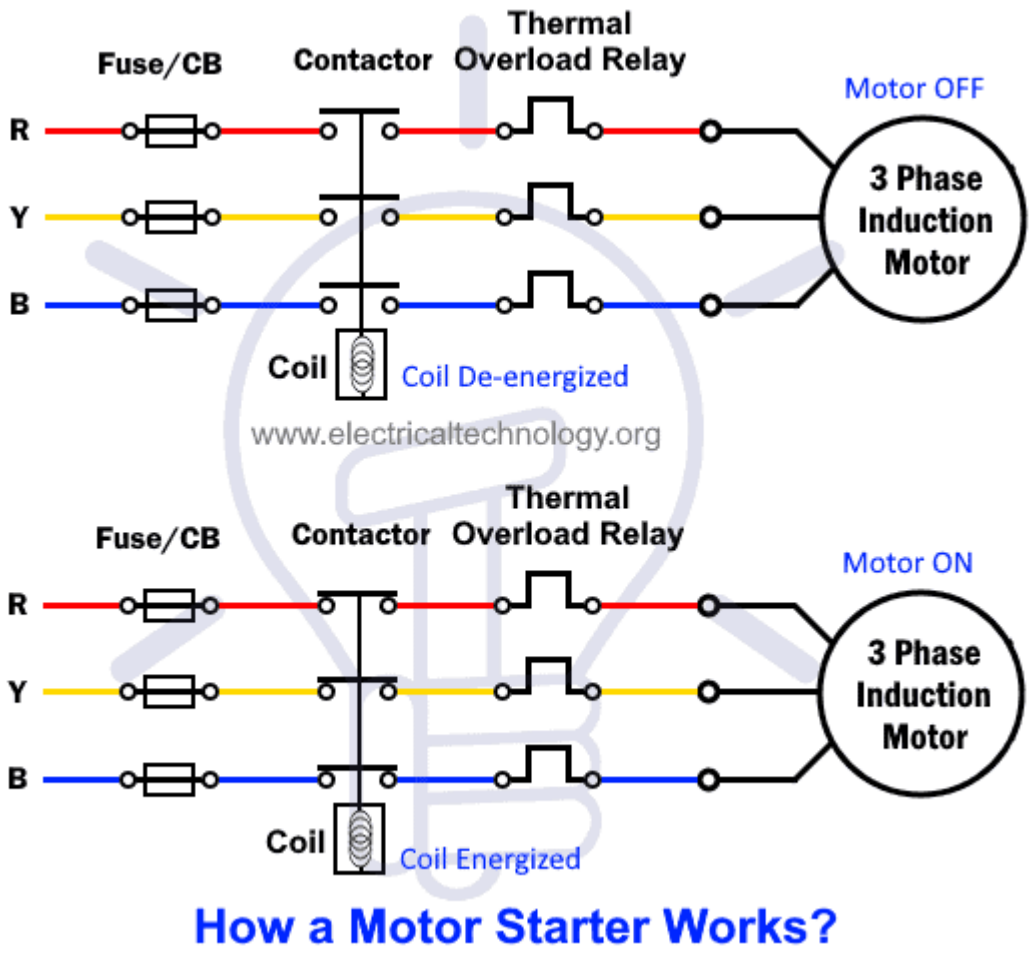
The manual starter is used for smaller motors where the hand operated lever is manually operated (move the contacts position) to the ON or OFF position. The disadvantage of these kinds of starters is that they need to switch ON after power frailer. In other words, they need manual control for each (ON or OFF) operation. Sometimes, this operation may leads to flow high currents in the motor winding which may burn the motor. This is why it is not recommended in most cases where other alternative motor starters with protection are used such as automatic starters.

Control circuits for automatic starters of DC and AC motors

An automatic starter consists of an overload relay and a contactor is used to switch the motor ON/OFF operation and protect the motor from overheating. When current passes through the contactor coils, it energizes and produces the electromagnetic field which pulls or pushes the contacts to make the connection of motor windings to the power supply.

The start and stop push buttons connected to the motor and starter can be used for ON and OFF operation of motors. The contactor coils can be de-energize by pushing the stop button which leads to de-energize the coil. This way, the contactor contacts move back due to spring arrangement to its normal position which leads to switch off the motor. In case of power failure or manual switch-off operation, the motor won't start automatically until we manually start the motor by pressing the "start push button".

Control circuits for automatic starters of DC and AC motors

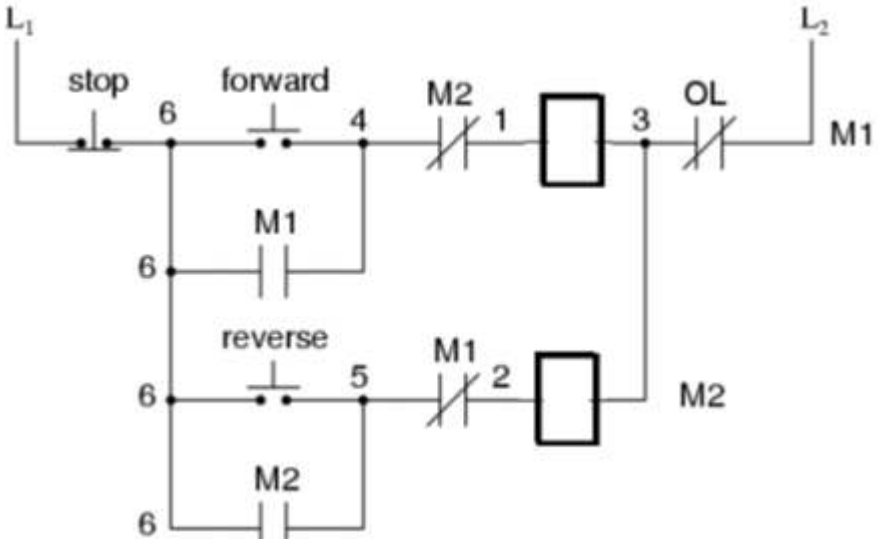


How a Motor Starter Works?

Control circuits for automatic starters of DC and AC motors

1-Control circuit of forward and reverse operation

- When the “Forward” pushbutton is actuated, M1 will energize, closing the normally-open auxiliary contact in parallel with that switch. When the pushbutton is released, the closed M1 auxiliary contact will maintain current to the coil of M1, thus latching the “Forward” circuit in the “on” state.
- The same sort of thing will happen when the “Reverse” pushbutton is pressed. These parallel auxiliary contacts are sometimes referred to as seal-in contacts, the word “seal” meaning essentially the same thing as the word latch.

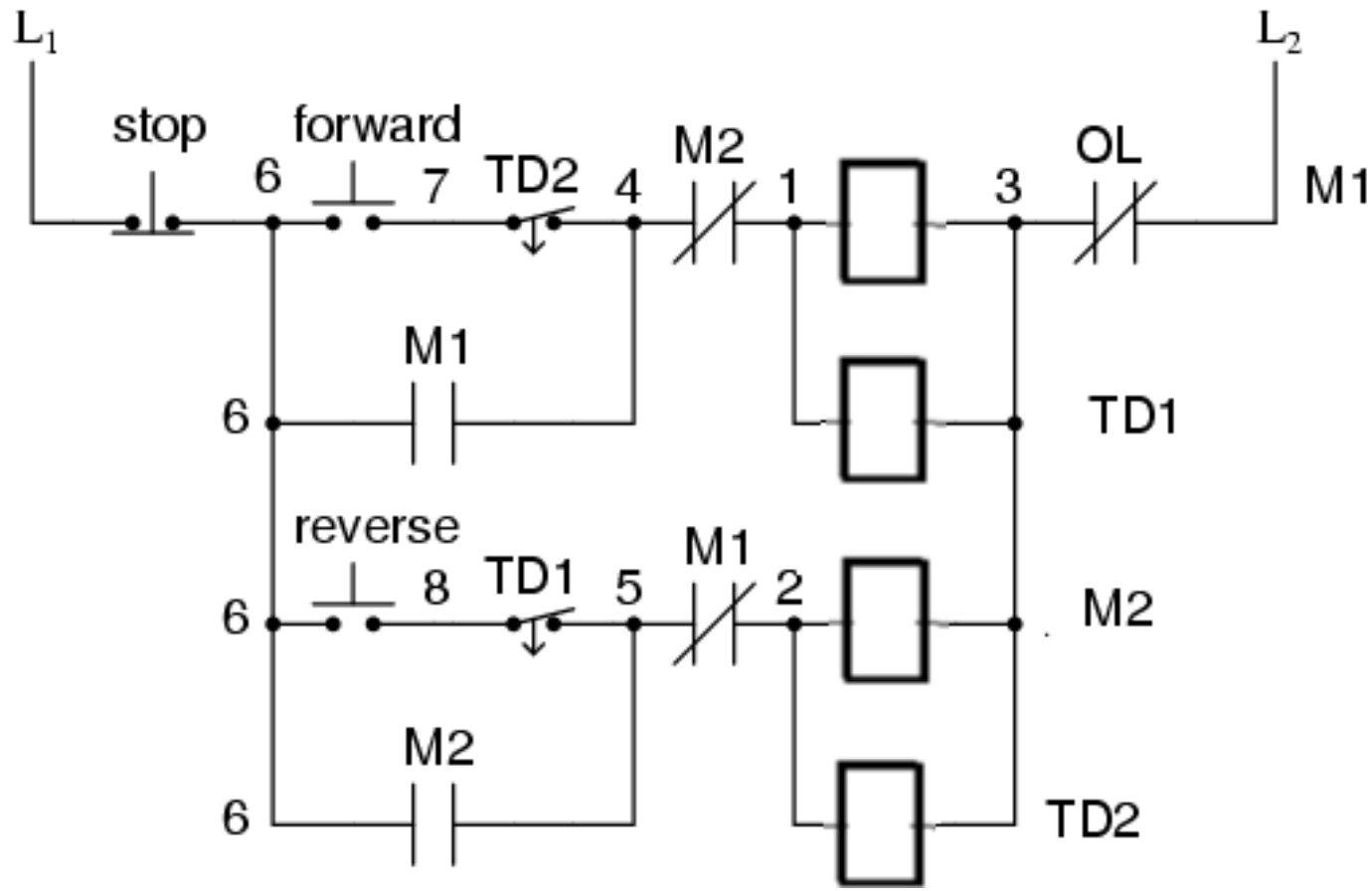


- if either forward or reverse circuits are latched, they may be

"unlatched" by momentarily pressing the "Stop" pushbutton, which will open either forward or reverse circuit, de-energizing the energized contactor, and returning the seal-in contact to its normal (open) state. The "Stop" switch, having normally-closed contacts, will conduct power to either forward or reverse circuits when released.

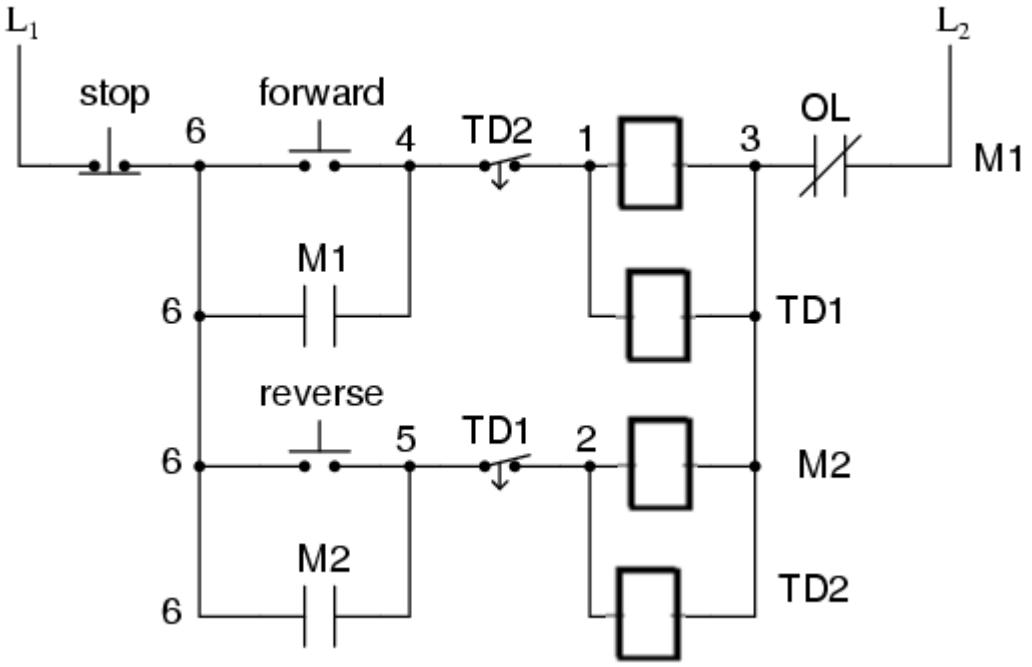
2. If our motor turned a mechanical load with a lot of momentum, such as a large air fan, the motor might continue to move for a substantial amount of time after the stop button had been pressed.

This could be problematic if an operator were to try to reverse the motor direction without waiting for the fan to stop turning.
If the fan was still moving forward and the “Reverse” pushbutton was pressed, the motor would struggle to overcome that inertia of the large fan as it tried to begin turning in reverse, drawing excessive current and potentially reducing the life of the motor, drive mechanisms, and fan. What we might like to have is some kind of a time-delay function in this motor control system to prevent such a premature (مبكر)startup from happening. Let’s begin by adding a couple of time-delay relay coils, one in parallel with each motor contactor coil. If we use contacts that delay returning to their normal state, these relays will provide us a “memory” of which direction the motor was last powered to turn. What we want each time-delay contact to do is to open the starting-switch leg of the opposite rotation circuit for several seconds, while the fan decelerates to a halt.


L1
L2
stop
forward
6
7
TD2
4
M2
1
3
OL
M1
M1
6
TD1
reverse
8
TD1
5
M1
2
6
M2
M2
6
TD2

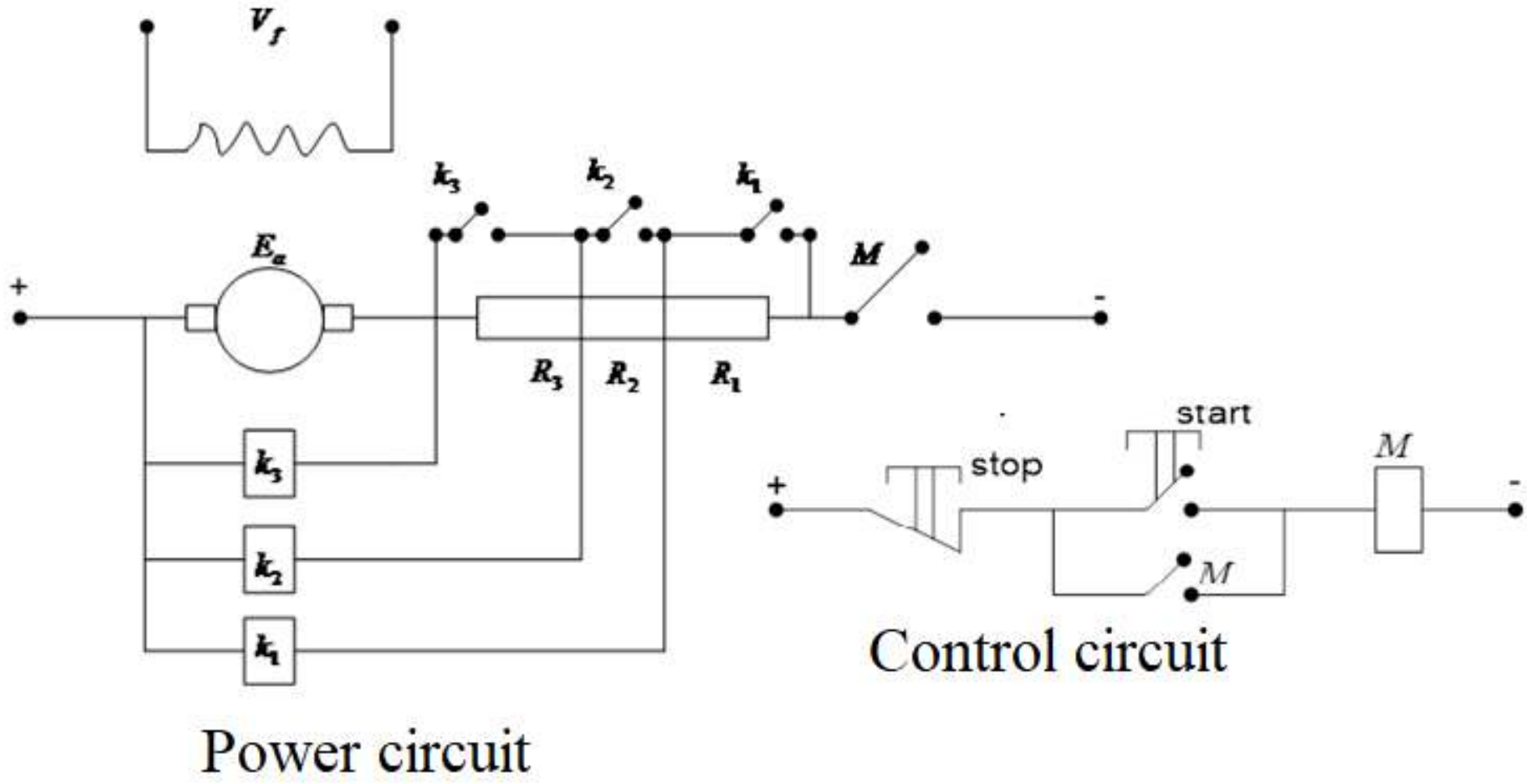
- We can get rid of auxiliary contacts M1 and M2 for interlocks and just use TD1 and TD2‘s contacts, since they immediately open when their respective relay coils are energized, thus “locking out” one contactor if the other is energized. Each time delay relay will serve a dual purpose: preventing the other contactor from energizing while

The motor is running and preventing the same contactor from energizing until a prescribed time after motor shutdown. The resulting circuit has the advantage of being simpler than the previous example



Automatic starting of DC motor

Starting as a function voltage (three- step starting)



For the previous figure:

$$E_{a1} = k_e \phi \omega_1$$

$$E_{a2} = k_e \phi \omega_2$$

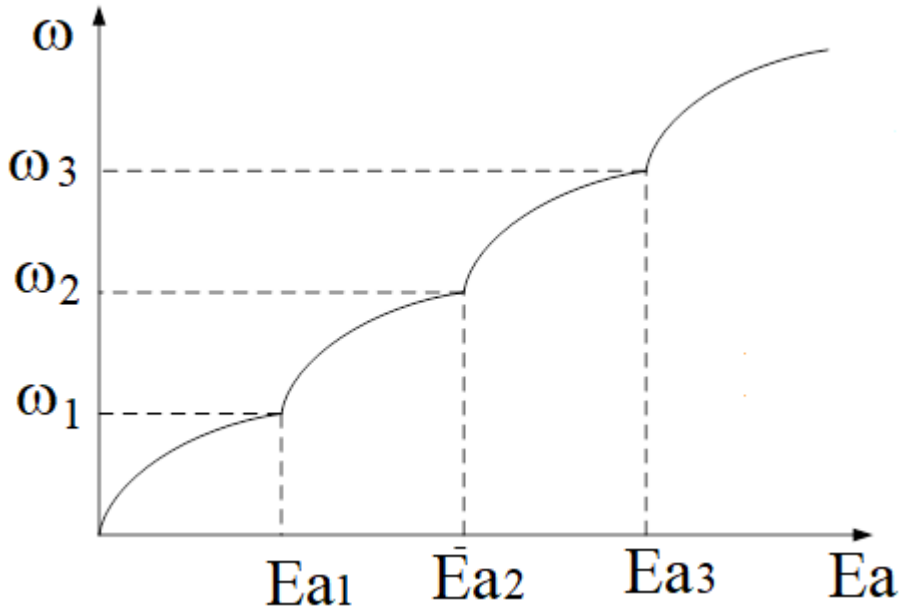
$$E_{a3} = k_e \phi \omega_3$$

$$V_{k1} = k\emptyset\omega_1 + I_a(R_a + R_2 + R_3)$$

$$V_{k2} = k\emptyset\omega_2 + I_a(R_a + R_3)$$

$$V_{k3} = k\emptyset\omega_3 + I_a(R_a)$$

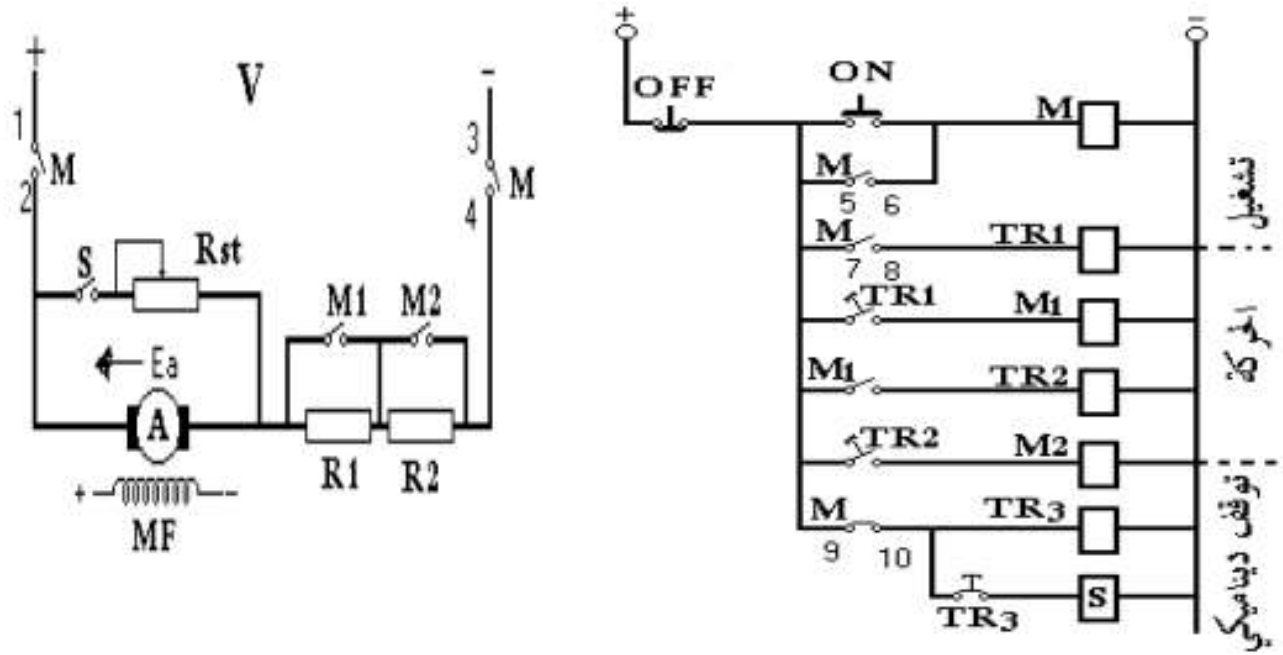
Where $V_{k1} = V_{k2} = V_{k3} = V_k$



Separately Excited DC Motor Starting in Two Steps and Dynamic Braking using Time Principle:

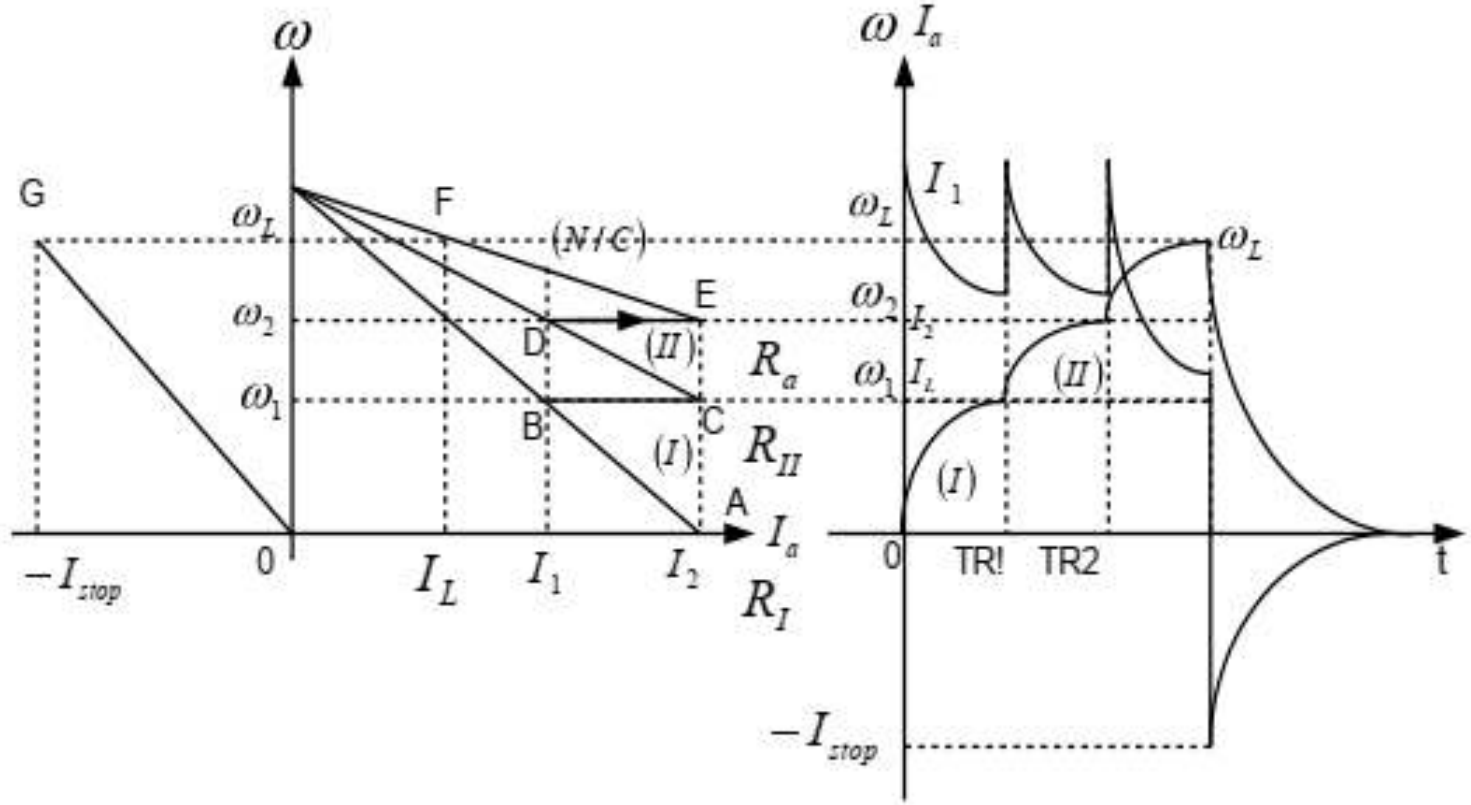
Elements of the electrical diagram:

M :Main contactor, M1: First step contactor, TR1: First step time delay ON timer
M2: Second step contactor, TR2: Second step time delay ON timer
S: -Stopping contactor, TR3: - Stopping time delay ON timer



A) Power circuit B) Control circuit

static and dynamic characteristics of a two-step starting and dynamic braking using time principle of separately excited dc motor.



A) Static characteristic **b) Dynamic characteristic**

AC motors automatic starting

- **Direct Starting and Mechanical Braking of AC Motors**
  Principle of operation :
  When the push button (ON) is pressed the contactor coil ((M) is activated, then the stator windings are connected to the source as shown in figure (9-1-A), and the contactor automatically feeds itself as in figure (9-1-B), so the motor is started on the natural torque-speed characteristic until it operates at operating steady-state point at load speed ω_L and the load torque T_L as shown in Figure (9-1-C). When the push button (OFF) is pressed the contactor coil ((M) deactivated and the motor is disconnected from the source and stops mechanically.
  Dynamic characteristics of direct starting and mechanical braking
  are shown in figure (9-1-D).

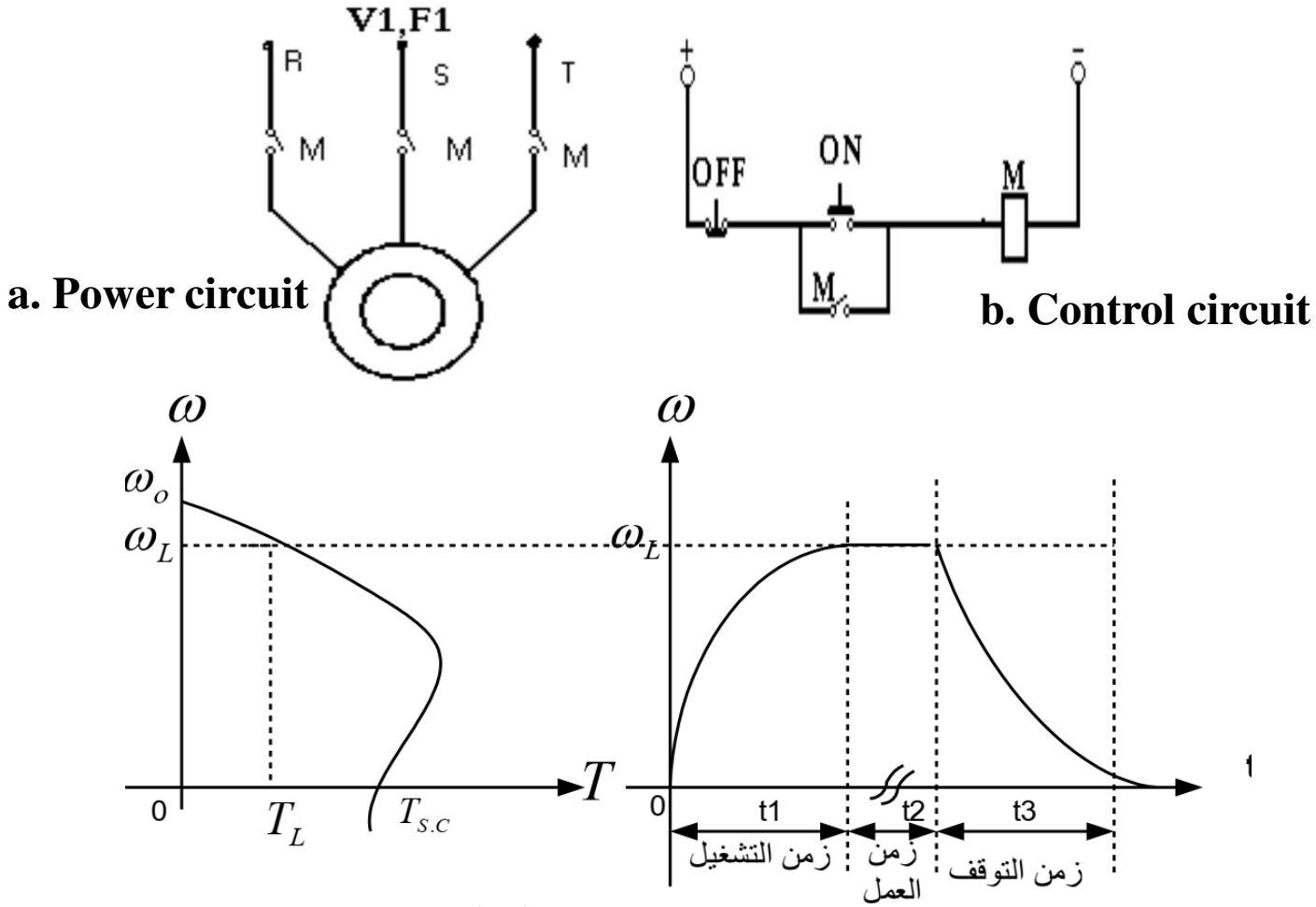


Figure (9-1) AC Motors direct starting and mechanical braking characteristics

Indirect ac motor starting

- Squirrel – cage motors are frequently started by connecting then directly across the supply lines. A large starting current of the order of 500 up to 700 percent of full load current may flow in the lines, if this causes appreciable voltage drop in the line, it may affect other drives connected to the line. Also, if a large current flows for a long time it may overheat the motor and damage the insulation. In such cases, indirect starting is achieved

1. A three – phase step down autotransformer, as shown in fig .1, may employ as a reduced – voltage starter. As a motor approaches certain speed, the transformer is switched out of the circuit

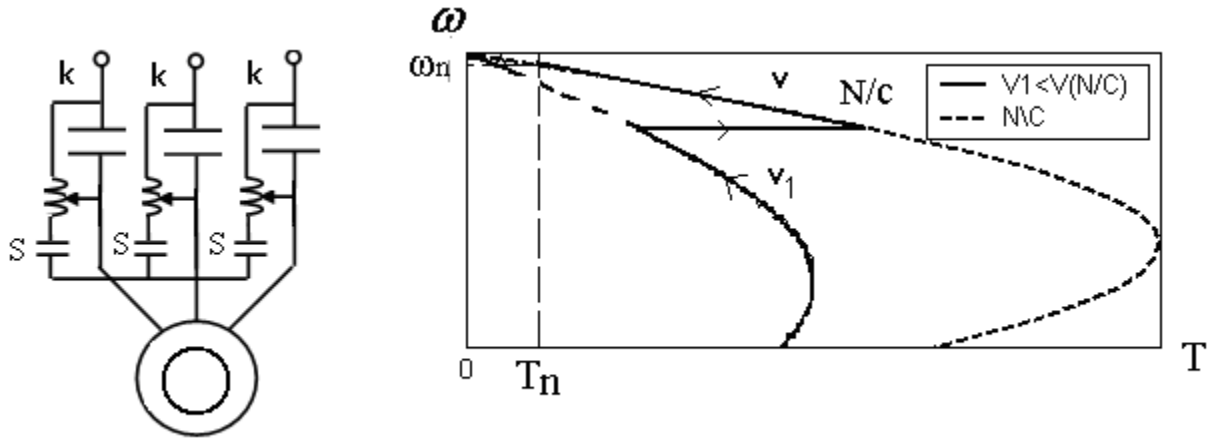
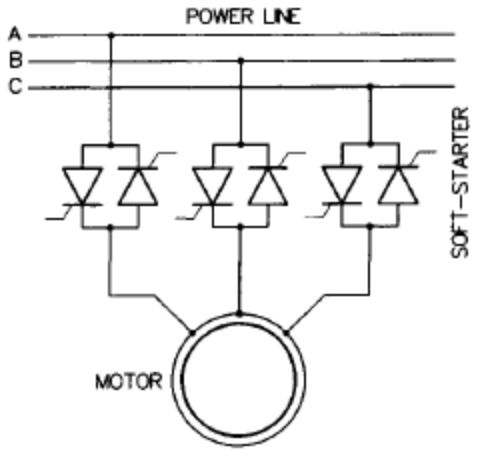


Fig.1

2. A solid-state voltage controller, as shown in Fig. 3, can also be used as a reduced–voltage starter. The controller can provide a smooth starting. These methods result in a decrease in the starting torque because the torque developed is proportional to the square of the terminal voltage.



3. *Impedance starting,* illustrated in the Figure 4, series impedances (resistive or reactive) are inserted between the power line and the motor to limit the starting current. As the motor gains speed, the impedances are shorted out, first by contacts 1, then by contacts 2. Also, starting can be done by inserting resistance into the rotor circuit as shown in fig.5 where its characteristics are shown in fig.6.

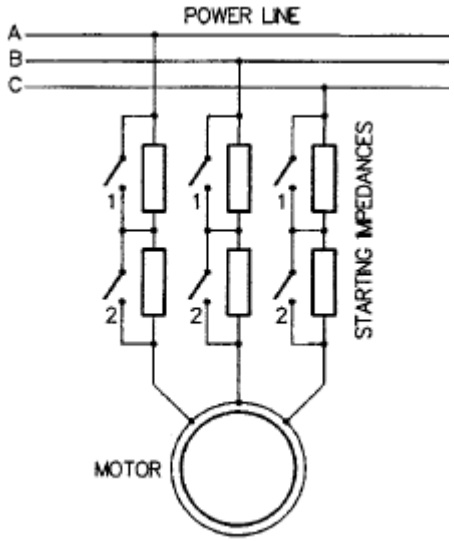
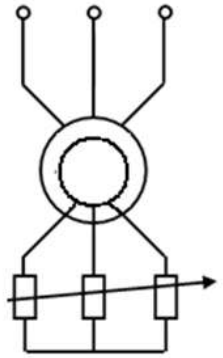
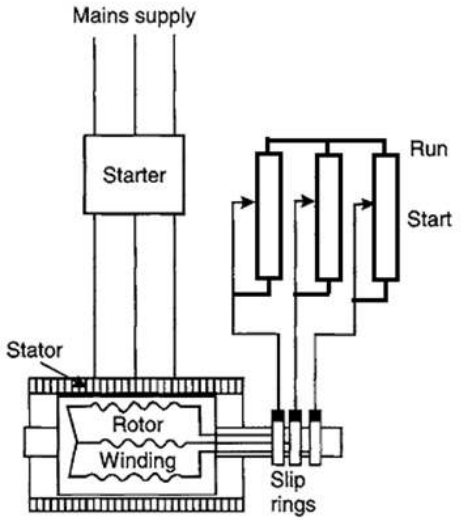


Figure 4



a)



b)

Fig.5

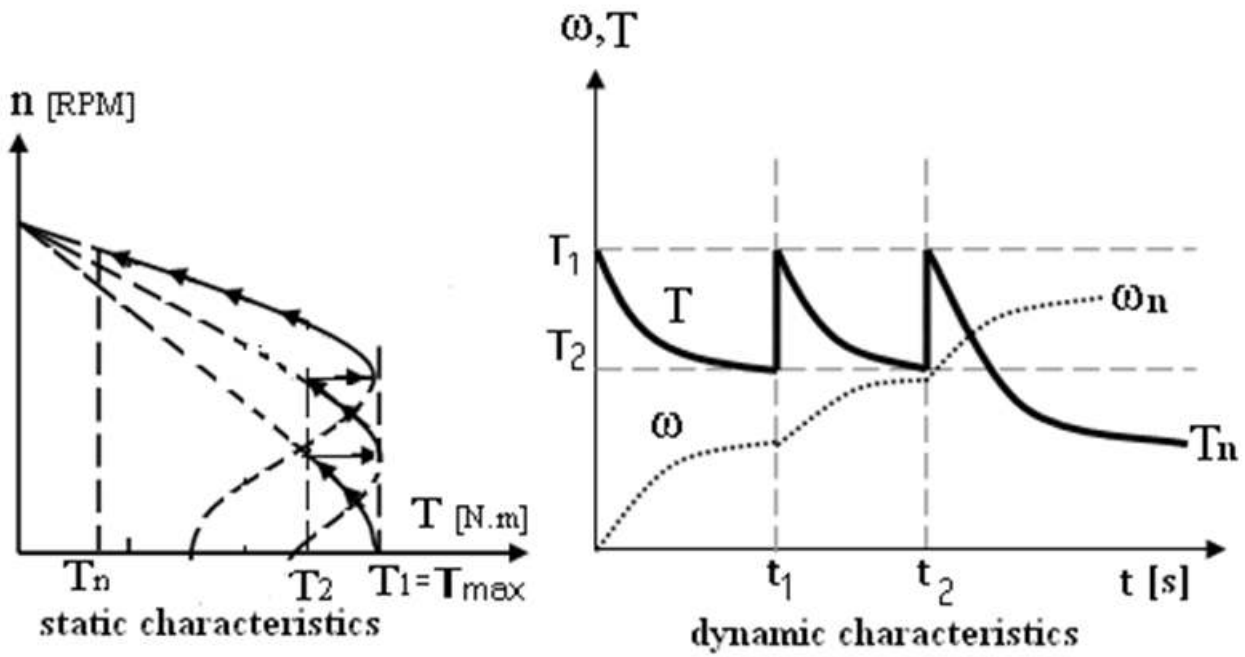
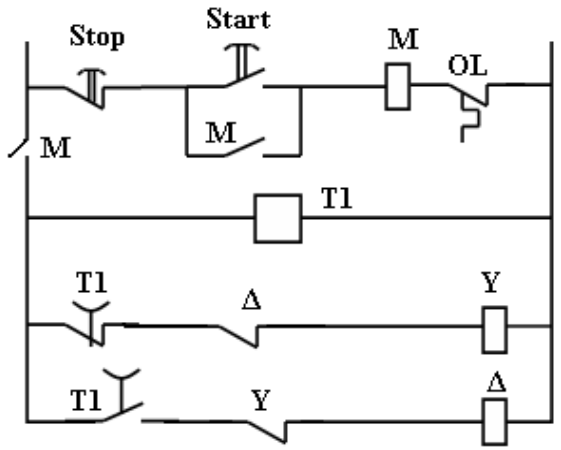
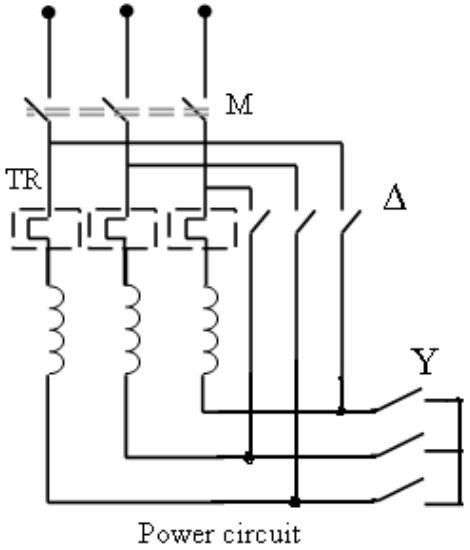

n [RPM]
T [N.m]
Tn
T2
T1=Tmax
static characteristics
a)
ω,T
T1
T2
T
ω
ωn
Tn
t1
t2
t [s]
dynamic characteristics
b)

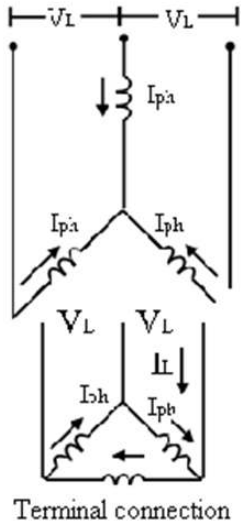
Fig.6

4. Automatic starting by γ-Δ reconnection:

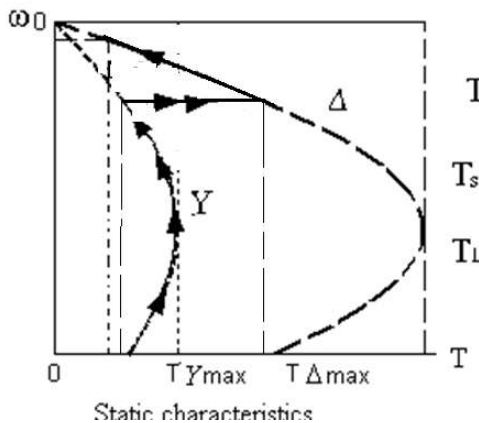




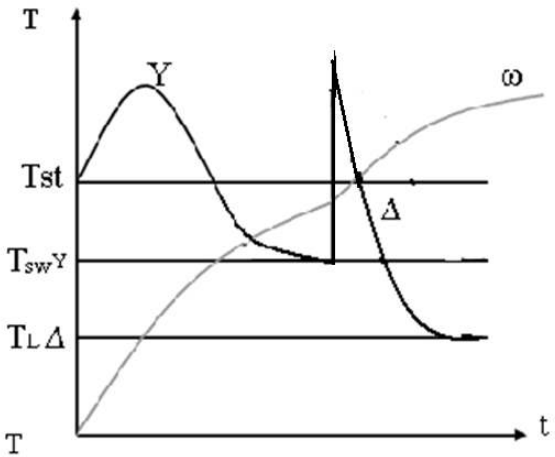
Power circuit



Terminal connection



Static characteristics



Dynamic characteristics